Section: Module 5

Instrumental Variable

Kentaro Nakamura

GOV 2002

October 10th, 2025

Agenda

- Two-Stage Least Squares (2SLS)
- Instrumental Variable (Binary Treatment)
  - Assumptions
  - Identification
- Multi-valued treatment
  - Identification
  - Point Estimate / Standard Error (Problem Set)
- Review questions

Motivation: Non-Compliance

- In experiment, sometimes we cannot directly force treatments
  - We instead give **encouragement** to take the treatment
  - But some units might refuse to take the treatment
- **Method 1**: Intention-to-Treat (ITT) analysis
  - Estimate the effect of encouragement, not treatment
- **Method 2**: Instrumental Variable
  - Use encouragement to identify the effect of treatment, not encouragement

Instrumental Variable: Linear Model

- Consider the following models:

$$Y_i = \alpha + \beta T_i + \epsilon_i$$
$$T_i = \gamma Z_i + \eta_i$$

  where $\mathbb{E}[\epsilon_i \mid Z_i] = \mathbb{E}[\eta_i \mid Z_i] = 0$

- But treatment is not randomized, so $\mathbb{E}[\epsilon_i \mid T_i] \neq 0$

- As a result, OLS estimator is biased and inconsistent

$$\hat{\beta}_{\text{OLS}} = \frac{\widehat{\text{Cov}}(Y_i, T_i)}{\widehat{\text{V}}(T_i)} = \beta + \frac{\widehat{\text{Cov}}(\epsilon_i, T_i)}{\widehat{\text{V}}(T_i)} \xrightarrow{p} \beta + \frac{\text{Cov}(\epsilon_i, \eta_i)}{\text{V}(T_i)} \neq \beta$$

Two Stage Least Squares

- Consider the following models:

$$Y_i = \alpha + \beta T_i + \epsilon_i$$
$$T_i = \gamma Z_i + \eta_i$$

  where $\mathbb{E}[\epsilon_i \mid Z_i] = \mathbb{E}[\eta_i \mid Z_i] = 0$

- **Wald Estimator**: $\hat{\beta}_{\text{IV}} := \frac{\widehat{\text{Cov}(Y_i, Z_i)}}{\widehat{\text{Cov}(T_i, Z_i)}} = \frac{\text{Effect of Z on Y}}{\text{Effect of Z on T}}$
- It works because

$$\beta_{\text{IV}} = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(T_i, Z_i)} = \frac{\text{Cov}(\alpha + \beta T_i + \epsilon, Z_i)}{\text{Cov}(\gamma Z_i + \eta_i, Z_i)} = \frac{\text{Cov}(\beta T_i, Z_i)}{\text{Cov}(\gamma Z_i, Z_i)}$$
$$= \frac{\text{Cov}(\beta(\gamma Z_i + \eta_i), Z_i)}{\text{Cov}(\gamma Z_i, Z_i)} = \frac{\beta\gamma\mathbb{V}[Z_i]}{\gamma\mathbb{V}[Z_i]} = \beta$$

- However, this assumes the linear model. $\rightarrow$ Can we derive Instrumental Variable without linear model?

Non-Compliance: Complier

- Four principal strata (or compliance types):
  1. Compliers: $T_i(Z_i = 1) = 1$ and $T_i(Z_i = 0) = 0$
  2. Always-takers: $T_i(Z_i = 1) = T_i(Z_i = 0) = 1$
  3. Never-takers: $T_i(Z_i = 1) = T_i(Z_i = 0) = 0$
  4. Defiers: $T_i(Z_i = 1) = 0$ and $T_i(Z_i = 0) = 1$

	$Z_i = 1$	$Z_i = 0$
$T_i = 1$	Complier / Always-taker	Defier / Always-taker
$T_i = 0$	Defier / Never-taker	Complier / Never-taker

Instrumental Variable

- **Assumption 1 (Randomization)**: Instruments are randomized so that

$$\{Y_i(T_i = t, Z_i = z), T_i(Z_i = z)\} \perp\!\!\!\perp Z_i$$

- **Assumption 2 (Exclusion Restriction)**: Instruments affect outcome only through treatment so that

$$Y_i(T_i = t, Z_i = z) = Y_i(Z_i = z)$$

- **Assumption 3 (Monotonicity)**: No defiers (i.e., those who would take treatment when not encouraged, but would not when encouraged)

$$T_i(Z_i = 1) \geq T_i(Z_i = 0)$$

- **Assumption 4 (Relevance)**: Encouragement has effect on treatment

$$\mathbb{E}[T_i(Z_i = 1) - T_i(Z_i = 0)] \neq 0$$

Instrumental Variable: Identification (1)

- ITT effect is defined as

$$\mathrm{ITT} := \mathbb{E}[Y_i(Z_i = 1) - Y_i(Z_i = 0)]$$

- By randomization and consistency, we have

$$\mathbb{E}[Y_i(Z_i = z)] = \mathbb{E}[Y_i(Z_i = z) \mid Z_i = z] = \mathbb{E}[Y_i \mid Z_i = z]$$

- Thus, ITT effect is identified as

$$\mathbb{E}[Y_i(Z_i = 1) - Y_i(Z_i = 0)] = \mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]$$

Instrumental Variable: Identification (2)

- Now, notice that

$$Y_i(Z_i = z) = Y_i(T_i = T_i(z), Z_i = z)$$

- This is because $T_i(z)$ is a value of treatment under $Z_i = z$
  - So, $Y_i(T_i = T_i(z), Z_i = z)$ is function of only $Z_i = z$

- So, we can write ITT Effect as

$$\begin{aligned}&\mathbb{E}[Y_i(Z_i = 1, T_i(Z_i = 1)) - Y_i(Z_i = 0, T_i(Z_i = 1))] \\ &= \underbrace{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}_{\text{Estimatable from Data (Identified)}}\end{aligned}$$

Instrumental Variable: Identification (3)

- By law of iterated Expectation, we can write

$$\mathbb{E}[Y_i(1, T_i(1)) - Y_i(0, T_i(0))] = \mathbb{E}\Big[\mathbb{E}[\Delta \mid \underbrace{T_i(1), T_i(0)}_{\text{Principal Strata}}]\Big]$$

  where $\Delta = Y_i(1, T_i(1)) - Y_i(0, T_i(0))$
- This is equivalent to

$$= \mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 0] \underbrace{\mathbb{P}(T_i(1) = 1, T_i(0) = 0)}_{\text{Complier}}$$

$$+ \mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 1] \underbrace{\mathbb{P}(T_i(1) = 1, T_i(0) = 1)}_{\text{Always-Taker}}$$

$$+ \mathbb{E}[\Delta \mid T_i(1) = 0, T_i(0) = 0] \underbrace{\mathbb{P}(T_i(1) = 0, T_i(0) = 0)}_{\text{Never-Taker}}$$

$$+ \mathbb{E}[\Delta \mid T_i(1) = 0, T_i(0) = 1] \underbrace{\mathbb{P}(T_i(1) = 0, T_i(0) = 1)}_{\text{Defier}}$$

Instrumental Variable: Identification (4)

- Notice that $\mathbb{P}(T_i(1) = 0, T_i(0) = 1) = 0$ (i.e., no defier)
- Also notice that for always taker,

$$\begin{aligned}
&\mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 1] \\
&= \mathbb{E}[Y_i(Z_i = 1, T_i(1)) - Y_i(Z_i = 0, T_i(0)) \mid T_i(1) = 1, T_i(0) = 1] \\
&= \mathbb{E}[Y_i(Z_i = 1, T_i = 1) - Y_i(Z_i = 0, T_i = 1) \mid T_i(1) = 1, T_i(0) = 1]
\end{aligned}$$

- By exclusion, we have $Y_i(T_i = t, Z_i = z) = Y_i(Z_i = z)$. So,

$$= \mathbb{E}[\underbrace{Y_i(T_i = 1) - Y_i(T_i = 1)}_{=0} \mid T_i(1) = 1, T_i(0) = 1] = 0$$

- Same holds for never-taker

Instrumental Variable: Identification (5)

- Hence,

$$
\begin{aligned}
&\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0] \\
&= \mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 0] \underbrace{\mathbb{P}(T_i(1) = 1, T_i(0) = 0)}_{\text{Complier}} \\
&\quad + \underbrace{\mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 1]}_{=0} \mathbb{P}(T_i(1) = 1, T_i(0) = 1) \\
&\quad + \underbrace{\mathbb{E}[\Delta \mid T_i(1) = 0, T_i(0) = 0]}_{=0} \mathbb{P}(T_i(1) = 0, T_i(0) = 0) \\
&\quad + \mathbb{E}[\Delta \mid T_i(1) = 0, T_i(0) = 1] \underbrace{\mathbb{P}(T_i(1) = 0, T_i(0) = 1)}_{=0} \\
&= \mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 0] \underbrace{\mathbb{P}(T_i(1) = 1, T_i(0) = 0)}_{\text{Complier}}
\end{aligned}
$$

Instrumental Variable: Identification (6)

- Finally, notice that

$$
\begin{aligned}
&\mathbb{E}[T_i(1) - T_i(0)] \\
&= \mathbb{E}[T_i(1) \mid Z_i = 1] - \mathbb{E}[T_i(0) \mid Z_i = 0] \quad (\because \text{Randomization}) \\
&= \mathbb{E}[T_i \mid Z_i = 1] - \mathbb{E}[T_i \mid Z_i = 0] \quad (\because \text{Consistency}) \\
&= \mathbb{P}(T_i = 1 \mid Z_i = 1) - \mathbb{P}(T_i = 1 \mid Z_i = 0) \quad (\because T_i \text{ is binary}) \\
&= \mathbb{P}(\text{Always-taker or Complier}) - \mathbb{P}(\text{Always-Taker}) \quad (\because \text{no defier}) \\
&= \mathbb{P}(\text{Complier}) = \mathbb{P}(T_i(1) = 1, T_i(0) = 0)
\end{aligned}
$$

- Therefore,

$$
\underbrace{\mathbb{E}[\Delta \mid T_i(1) = 1, T_i(0) = 0]}_{\text{Late (ATE among complier)}} = \underbrace{\frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[T_i \mid Z_i = 1] - \mathbb{E}[T_i \mid Z_i = 0]}}_{\text{Estimatable from Data}}
$$

Estimating Complier for multi-valued treatment (1)

- What if we have multi-valued treatment
  - This is the setting of Question 3
- Consider the case where treatment is three category
  - I.e., $T_i \in \{0, 1, 2\}$
  - We keep instrument binary: $Z_i \in \{0, 1\}$
- The big change is that we have more principal strata
- How many principal strata do we have?

Estimating Complier for multi-valued treatment (2)

- We have 9 principal strata
  1. $\{T_i(Z_i = 0) = 0, T_i(Z_i = 1) = 0\}$
  2. $\{T_i(Z_i = 0) = 0, T_i(Z_i = 1) = 1\}$
  3. $\{T_i(Z_i = 0) = 0, T_i(Z_i = 1) = 2\}$
  4. $\{T_i(Z_i = 0) = 1, T_i(Z_i = 1) = 0\}$
  5. $\{T_i(Z_i = 0) = 1, T_i(Z_i = 1) = 1\}$
  6. $\{T_i(Z_i = 0) = 1, T_i(Z_i = 1) = 2\}$
  7. $\{T_i(Z_i = 0) = 2, T_i(Z_i = 1) = 0\}$
  8. $\{T_i(Z_i = 0) = 2, T_i(Z_i = 1) = 1\}$
  9. $\{T_i(Z_i = 0) = 2, T_i(Z_i = 1) = 2\}$

- We need to remove some of principal strata to identify the probability

Estimating Complier for multi-valued treatment (2)

- Monotonicity: $T_i(1) \geq T_i(0)$
  - Remove strata with $T_i(0) > T_i(1)$

	$T_i(1)$		
$T_i(0)$	0	1	2
0	$(0,0)$	$(0,1)$	$(0,2)$
1	$\times$	$(1,1)$	$(1,2)$
2	$\times$	$\times$	$(2,2)$

- We have 6 principal strata (remove 3 strata)
- These probabilities are not separately identified

Why 6 principal strata are not identified

$T_i(0)/T_i(1)$	0	1	2
0	p_{00}	p_{01}	p_{02}
1	$\times$	p_{11}	p_{12}
2	$\times$	$\times$	p_{22}

- We have 6 unknown probabilities ($p_{00},\ p_{01},\ p_{02},\ p_{11},\ p_{12},\ p_{22}$), but they must sum to one, so there are only **5 free parameters**.
- From the data, we observe the marginals for $k = 0, 1, 2$.

$$\mathbb{P}(T_i(0) = k) = \mathbb{P}(T_i = k \mid Z_i = 0), \quad \mathbb{P}(T_i(1) = k) = \mathbb{P}(T_i = k \mid Z_i = 1),$$

- So we observe 6 marginal probabilities, but they satisfy

$$\sum_{k=0}^{2} \mathbb{P}(T_i(0) = k) = 1, \qquad \sum_{k=0}^{2} \mathbb{P}(T_i(1) = k) = 1,$$

so they contain only 4 independent pieces of information.

Estimating Complier for multi-valued treatment (3)

- Another example: $T_i(1) - T_i(0) \in \{-1, 0\}$
  - Treatment can decrease by at most one

	$T_i(1)$		
$T_i(0)$	0	1	2
0	$(0,0)$	$\times$	$\times$
1	$(1,0)$	$(1,1)$	$\times$
2	$\times$	$(2,1)$	$(2,2)$

- We have 5 principal strata (remove 4 strata)
- In this case, we can identify the proportion of each stratum (we will show below)

Estimating Complier for multi-valued treatment (4)

- Can we estimate the proportion of each strata with standard errors?
- Because Z_i is randomized, for $k \in \{0, 1, 2\}$

$$\mathbb{P}(T_i(1) = k) = \mathbb{P}(T_i(1) = k \mid Z_i = 1) = \mathbb{P}(T_i = k \mid Z_i = 1)$$
$$\mathbb{P}(T_i(0) = k) = \mathbb{P}(T_i(0) = k \mid Z_i = 0) = \mathbb{P}(T_i = k \mid Z_i = 0)$$

- Moreover, because sum of probability is 1,

$$\begin{aligned}
\mathbb{P}(T_i(1) = 2 \mid T_i(0) = 2) + \mathbb{P}(T_i(1) = 1 \mid T_i(0) = 2) &= 1 \\
\mathbb{P}(T_i(1) = 1 \mid T_i(0) = 1) + \mathbb{P}(T_i(1) = 0 \mid T_i(0) = 1) &= 1 \\
\mathbb{P}(T_i(1) = 0 \mid T_i(0) = 0) &= 1
\end{aligned}$$

Estimating Complier for multi-valued treatment (5)

- Now, notice that by the definition of conditional probability,

$$1 = \mathbb{P}(T_i(1) = 0 \mid T_i(0) = 0) = \frac{\mathbb{P}(T_i(1) = 0, T_i(0) = 0)}{\mathbb{P}(T_i(0) = 0)}$$

- This means that

$$\mathbb{P}(T_i(0) = 0) = \mathbb{P}(T_i(1) = 0, T_i(0) = 0)$$

- Hence,

$$\mathbb{P}(T = 0 \mid Z = 0) = \mathbb{P}(T_i(1) = 0, T_i(0) = 0)$$

Estimating Complier for multi-valued treatment (6)

- Now, notice that

$$\mathbb{P}(T_i(1) = 0) = \mathbb{P}(T_i(0) = 1, T_i(1) = 0) + \mathbb{P}(T_i(0) = 0, T_i(1) = 0)$$

- We know $\mathbb{P}(T_i(1) = 0) = \mathbb{P}(T_i = 0 \mid Z_i = 1)$
- We also identify $\mathbb{P}(T_i(0) = 0, T_i(1) = 0) = \mathbb{P}(T = 0 \mid Z = 0)$ from previous step
- Therefore,

$$\mathbb{P}(T_i(0) = 1, T_i(1) = 0) = \mathbb{P}(T_i = 0 \mid Z_i = 1) - \mathbb{P}(T_i = 0 \mid Z_i = 0)$$

Estimating Complier for multi-valued treatment (7)

- What about $\mathbb{P}(T_i(0) = 1, T_i(1) = 1)$?

$$\mathbb{P}(T_i(0) = 1, T_i(1) = 1) = \mathbb{P}(T_i(0) = 1) - \mathbb{P}(T_i(0) = 1, T_i(1) = 0)$$

- As we identify $\mathbb{P}(T_i(0) = 1, T_i(1) = 0)$ already,

$$\begin{aligned}&\mathbb{P}(T_i(0) = 1, T_i(1) = 1) \\ &= \mathbb{P}(T_i = 1 \mid Z_i = 0) - (\mathbb{P}(T_i = 0 \mid Z_i = 1) - \mathbb{P}(T_i = 0 \mid Z_i = 0))\end{aligned}$$

- We simply repeat the same thing to identify all the strata

Estimating Complier for multi-valued treatment (8)

- Can you estimate the standard error?

$$\mathbb{P}(\widehat{T_i(0) = 1, T_i(1) = 0}) = \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1}) - \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0})$$

- Recall that $\mathbb{V}[X - Y] = \mathbb{V}[X] + \mathbb{V}[Y] - 2\mathrm{Cov}(X, Y)$
- Hence,

$$\begin{aligned}
&\mathbb{V}\Big(\mathbb{P}(\widehat{T_i(0) = 1, T_i(1) = 0})\Big) \\
&= \mathbb{V}\Big(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1}) - \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0})\Big) \\
&= \mathbb{V}\Big(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1})\Big) + \mathbb{V}\Big(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0})\Big) \\
&\quad - 2 \underbrace{\mathrm{Cov}\Big(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1}), \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0})\Big)}_{=0(\because \text{Independence})}
\end{aligned}$$

Estimating Complier for multi-valued treatment (9)

- What about $\mathbb{P}(T_i(0) = 1, T_i(1) = 1)$?

$$\begin{aligned}
&\widehat{\mathbb{P}(T_i(0) = 1, T_i(1) = 1)} \\
&= \widehat{\mathbb{P}(T_i = 1 \mid Z_i = 0)} - \left(\widehat{\mathbb{P}(T_i = 0 \mid Z_i = 1)} - \widehat{\mathbb{P}(T_i = 0 \mid Z_i = 0)}\right)
\end{aligned}$$

- Hence,

$$\begin{aligned}
&\mathbb{V}\left(\widehat{\mathbb{P}(T_i(0) = 1, T_i(1) = 1)}\right) \\
&= \mathbb{V}\left(\widehat{\mathbb{P}(T_i = 1 \mid Z_i = 0)} - \left(\widehat{\mathbb{P}(T_i = 0 \mid Z_i = 1)} - \widehat{\mathbb{P}(T_i = 0 \mid Z_i = 0)}\right)\right)
\end{aligned}$$

- **Caution**: Notice that $\widehat{\mathbb{P}(T_i = 1 \mid Z_i = 0)}$ and $\widehat{\mathbb{P}(T_i = 0 \mid Z_i = 0)}$ are not independent!
  - They are on the same sample ($Z_i = 0$)
  - Thus, you need to take into account **covariance**

Estimating Complier for multi-valued treatment (9)

$$
\begin{aligned}
&\mathbb{V}\Big(\mathbb{P}(\widehat{T_i = 1 \mid Z_i = 0}) - (\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1}) - \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0}))\Big) \\
&= \mathbb{V}(\mathbb{P}(\widehat{T_i = 1 \mid Z_i = 0})) \\
&\quad + \mathbb{V}(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1})) \\
&\quad + \mathbb{V}(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0}))) \\
&\quad - 2\underbrace{\mathrm{Cov}(\mathbb{P}(\widehat{T_i = 1 \mid Z_i = 0}), \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1}))}_{=0} \\
&\quad + 2\underbrace{\mathrm{Cov}(\mathbb{P}(\widehat{T_i = 1 \mid Z_i = 0}), \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0}))}_{\text{Non-Zero!}} \\
&\quad - 2\underbrace{\mathrm{Cov}(\mathbb{P}(\widehat{T_i = 0 \mid Z_i = 1}), \mathbb{P}(\widehat{T_i = 0 \mid Z_i = 0}))}_{=0}
\end{aligned}
$$

Estimating Complier for multi-valued treatment (10)

- How can we calculate covariance term?

$$\mathrm{Cov}(\mathbb{P}(T_i = \widehat{1 \mid Z_i = 0}), \mathbb{P}(T_i = \widehat{0 \mid Z_i = 0}))$$

- Why is it non-zero? $\rightarrow$ If $\mathbb{P}(T_i = \widehat{1 \mid Z_i = 0})$ becomes larger, $\mathbb{P}(T_i = \widehat{0 \mid Z_i = 0})$ should be smaller

- From theory of multinomial distribution:

$$\begin{aligned}
&\mathrm{Cov}(\mathbb{P}(T_i = \widehat{1 \mid Z_i = 0}), \mathbb{P}(T_i = \widehat{0 \mid Z_i = 0})) \\
&= -\frac{\mathbb{P}(T_i = 1 \mid Z_i = 0)\mathbb{P}(T_i = 0 \mid Z_i = 0)}{\text{Number of } Z_i = 0}
\end{aligned}$$

- Same for other covariance terms

What is multinominal distribution?

- Suppose there are K categories
- Think about how many times the trial where each T_i falls into one of K categories with probability

$$\mathbb{P}(T_i = k) = p_k \quad \text{with} \quad \sum_{k=1}^{K} p_k = 1$$

- We care about the counts of each category: i.e., $X_k = \sum_{i=1}^{n} \mathbb{1}\{T_i = k\}$
- The joint distribution of counts $(X_1, \cdots, X_K)$ follows multinominal distribution with $\sum_{k=1}^{K} X_k = n$
  - $\mathbb{E}[X_k] = np_k$
  - $\mathbb{V}[X_k] = np_k(1 - p_k)$
  - $\mathrm{Cov}(X_k, X_l) = -np_k p_l$
- As a result, the covariance of each probability estimates is

$$\mathrm{Cov}(\hat{p}_k, \hat{p}_l) = \mathrm{Cov}\left(\frac{X_k}{n}, \frac{X_l}{n}\right) = \frac{1}{n^2}\mathrm{Cov}(X_k, X_l) = -\frac{p_k p_l}{n}$$

Derivation: Multi-valued Treatment (1)

- Suppose that

$$T_i \in \{0, 1, \ldots, K-1\}, \qquad Z_i \in \{0, 1\}.$$

- Assume the followings (same as before)
  1. **Randomization**: $\{Y_i(0), Y_i(1), T_i(0), T_i(1)\} \perp\!\!\!\perp Z_i$.
  2. **Monotonicity**: $T_i(1) \geq T_i(0)$.
  3. **Exclusion restriction**: for each treatment level t, $Y_i(1, t) = Y_i(0, t)$.
- As we have seen, with monotonicity, the principal strata is not fully identified, and the 2SLS converges to

$$\widehat{\beta}_{\text{TSLS}} \xrightarrow{p} \beta_{\text{IV}} = \sum_{k=0}^{K-2} \sum_{j=k+1}^{K-1} w_{jk} \, \mathbb{E}\left[\frac{Y_i(1) - Y_i(0)}{j-k} \middle| \, T_i(1) = j, T_i(0) = k \right].$$

  - This is a weighted average of the causal effect for each principal stratum
  - Let's derive this result step by step.

Multi-valued Treatment: Identification (2)

- First, let $p := \mathbb{P}(Z_i = 1)$. Since Z_i is binary,

$$\begin{aligned}\mathbb{E}[Z_i] &= 0 \times (1-p) + 1 \times p = p, \\ \mathbb{E}[Y_i Z_i] &= 0 \times \mathbb{E}[Y_i \mid Z_i = 0](1-p) + 1 \times \mathbb{E}[Y_i \mid Z_i = 1]p \\ &= p\mathbb{E}[Y_i \mid Z_i = 1].\end{aligned}$$

- By the law of iterated expectation,

$$\mathbb{E}[Y_i] = p\mathbb{E}[Y_i \mid Z_i = 1] + (1-p)\mathbb{E}[Y_i \mid Z_i = 0].$$

- Substituting these expressions into the covariance,

$$\begin{aligned}&\mathrm{Cov}(Y_i, Z_i) = \mathbb{E}[Y_i Z_i] - \mathbb{E}[Y_i]\mathbb{E}[Z_i] \\ &= p\mathbb{E}[Y_i \mid Z_i = 1] - p\mathbb{E}[Y_i] \\ &= p\mathbb{E}[Y_i \mid Z_i = 1] - p\{p\mathbb{E}[Y_i \mid Z_i = 1] + (1-p)\mathbb{E}[Y_i \mid Z_i = 0]\} \\ &= p(1-p)\mathbb{E}[Y_i \mid Z_i = 1] - p(1-p)\mathbb{E}[Y_i \mid Z_i = 0] \\ &= p(1-p)\{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]\}.\end{aligned}$$

Multi-valued Treatment: Identification (3)

- The same calculation for treatment gives
$$\mathrm{Cov}(T_i, Z_i) = p(1-p)\{\mathbb{E}[T_i \mid Z_i = 1] - \mathbb{E}[T_i \mid Z_i = 0]\}.$$

- Hence, the common factor $p(1-p)$ cancels:
$$\beta_{\mathrm{IV}} := \frac{\mathrm{Cov}(Y_i, Z_i)}{\mathrm{Cov}(T_i, Z_i)} = \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[T_i \mid Z_i = 1] - \mathbb{E}[T_i \mid Z_i = 0]}.$$
- Since Z_i is binary and randomized,
$$\beta_{\mathrm{IV}} = \frac{\mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]}{\mathbb{E}[T_i(1)] - \mathbb{E}[T_i(0)]} = \frac{\mathbb{E}[Y_i(1) - Y_i(0)]}{\mathbb{E}[T_i(1) - T_i(0)]}.$$
- For TSLS with an intercept and no covariates, i.i.d. sampling and finite second moments imply
$$\widehat{\beta}_{\mathrm{TSLS}} = \frac{\widehat{\mathrm{Cov}}(Y_i, Z_i)}{\widehat{\mathrm{Cov}}(T_i, Z_i)} \xrightarrow{p} \beta_{\mathrm{IV}}.$$
- Let's analyze what β_{IV} is.

Multi-valued Treatment: Identification (4)

- Let $\Delta := Y_i(1) - Y_i(0)$ and $p_{jk} := \mathbb{P}(T_i(1) = j, T_i(0) = k)$.
- By the law of iterated expectation,

$$\mathbb{E}[\Delta] = \sum_{k=0}^{K-1} \sum_{j=0}^{K-1} \mathbb{E}[\Delta \mid T_i(1) = j, T_i(0) = k] p_{jk}.$$

- By monotonivity $p_{jk} = \mathbb{P}(T_i(1) = j, T_i(0) = k) = 0$. In addition, by exclusion restriction, $\Delta = 0$ for $j = k$. So, So,

$$\begin{aligned}\mathbb{E}[\Delta] &= \sum_{k=0}^{K-1} \sum_{j=0}^{K-1} \mathbb{E}[\Delta \mid T_i(1) = j, T_i(0) = k] p_{jk} \\ &= \sum_{k=0}^{K-2} \sum_{j=k+1}^{K-1} \mathbb{E}[\Delta \mid T_i(1) = j, T_i(0) = k] p_{jk}.\end{aligned}$$

Multi-valued Treatment: Identification (5)

- Substituting the numerator and denominator,

$$\beta_{\text{IV}} = \frac{\mathbb{E}[\Delta]}{\mathbb{E}[T_i(1) - T_i(0)]} = \sum_{k=0}^{K-2} \sum_{j=k+1}^{K-1} \frac{\mathbb{E}[\Delta \mid T_i(1) = j, T_i(0) = k] p_{jk}}{\mathbb{E}[T_i(1) - T_i(0)]}.$$

- For each affected stratum, multiply and divide by $j - k > 0$:

$$\frac{p_{jk}}{\mathbb{E}[T_i(1) - T_i(0)]} \mathbb{E}[\Delta \mid T_i(1) = j, T_i(0) = k]$$

$$= \frac{(j-k)p_{jk}}{\mathbb{E}[T_i(1) - T_i(0)]} \frac{\mathbb{E}[\Delta \mid T_i(1) = j, T_i(0) = k]}{j-k}$$

$$= \underbrace{\frac{(j-k)p_{jk}}{\mathbb{E}[T_i(1) - T_i(0)]}}_{:= w_{jk}} \mathbb{E}\left[\frac{\Delta}{j-k} \middle| T_i(1) = j, T_i(0) = k \right].$$

  - Conditional on the stratum, $j - k$ is a constant.

Multi-valued Treatment: Identification (6)

- Now, we analyze the weight. The denominator is written as

$$
\begin{aligned}
&\mathbb{E}[T_i(1) - T_i(0)] \\
&= \sum_{k=0}^{K-1} \sum_{j=0}^{K-1} \mathbb{E}[T_i(1) - T_i(0) \mid T_i(1) = j, T_i(0) = k] p_{jk} \\
&= \sum_{k=0}^{K-1} \sum_{j=0}^{K-1} (j-k) p_{jk} \\
&= \sum_{k=0}^{K-1} \sum_{j=k}^{K-1} (j-k) p_{jk} \qquad (\because \text{ Monotonicity}) \\
&= \sum_{k=0}^{K-2} \sum_{j=k+1}^{K-1} (j-k) p_{jk} \qquad (\because \ j-k=0 \text{ if } j=k).
\end{aligned}
$$

  - Within a stratum, the change in treatment is exactly $j-k$.

Multi-valued Treatment: Identification (7)

- Therefore, the weight of stratum (j, k) is written as

$$
\begin{aligned}
w_{jk} &:= \frac{(j-k)p_{jk}}{\mathbb{E}[T_i(1) - T_i(0)]} \\
&= \frac{(j-k)p_{jk}}{\sum_{k'=0}^{K-2} \sum_{j'=k'+1}^{K-1} (j'-k')p_{j'k'}} \\
&= \frac{(j-k)\mathbb{P}(T_i(1) = j, T_i(0) = k)}{\sum_{k'=0}^{K-2} \sum_{j'=k'+1}^{K-1} (j'-k')\mathbb{P}(T_i(1) = j', T_i(0) = k')}.
\end{aligned}
$$

- The primed indices sum over all affected strata.
- The numerator concerns the particular stratum (j, k).

Multi-valued Treatment: Identification (8)

- Therefore, we finally get

$$\widehat{\beta}_{\mathrm{TSLS}} \xrightarrow{p} \beta_{\mathrm{IV}} = \sum_{k=0}^{K-2} \sum_{j=k+1}^{K-1} w_{jk} \underbrace{\mathbb{E}\left[\frac{Y_i(1) - Y_i(0)}{j-k} \middle| T_i(1) = j, T_i(0) = k \right]}_{\text{Effect per-dose within a stratum}}.$$

- The weight is proportional to

$$w_{jk} \propto \underbrace{(j-k)}_{\text{Change in treatment}} \underbrace{\mathbb{P}(T_i(1) = j, T_i(0) = k)}_{\text{Stratum proportion}}.$$

- As we have seen, with monotonicity, the stratum proportion is not identified!

Two Stage Least Squares: Matrix Notation

- Consider the following model:

$$Y_i = \boldsymbol{X}_i^\top \beta + \epsilon_i, \qquad \boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{\epsilon}$$

  where $\boldsymbol{X}_i$ is $K \times 1$ and $\boldsymbol{Z}_i$ is $L \times 1$

- Stack the observations into matrices:

$$\underbrace{\boldsymbol{Y}, \boldsymbol{\epsilon}}_{n \times 1}, \qquad \underbrace{\boldsymbol{X}}_{n \times K}, \qquad \underbrace{\boldsymbol{Z}}_{n \times L}$$

  Row i of $\boldsymbol{X}$ is $\boldsymbol{X}_i^\top$, and row i of $\boldsymbol{Z}$ is $\boldsymbol{Z}_i^\top$

- Include the intercept and any exogenous covariates in both $\boldsymbol{X}$ and $\boldsymbol{Z}$
- Assume $\text{rank}(\boldsymbol{Z}) = L$ and $\text{rank}(\boldsymbol{Z}^\top \boldsymbol{X}) = K$, so $L \geq K$

Two Stage Least Squares: First Stage

- Regress each column of $\boldsymbol{X}$ on $\boldsymbol{Z}$. The OLS coefficient estimates are

$$(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{X}$$

- Multiply by $\boldsymbol{Z}$ to obtain the fitted values:

$$\widehat{\boldsymbol{X}} = \boldsymbol{Z} \underbrace{(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top \boldsymbol{X}}_{\text{OLS coefficient estimates}} = \underbrace{\boldsymbol{Z}(\boldsymbol{Z}^\top \boldsymbol{Z})^{-1} \boldsymbol{Z}^\top}_{\boldsymbol{P}_Z} \boldsymbol{X} = \boldsymbol{P}_Z \boldsymbol{X}$$

  where $\boldsymbol{P}_Z$ projects onto the column space of $\boldsymbol{Z}$

  - Recall that projection matrices are symmetric and idempotent (by definition)
  - In other words, $\boldsymbol{P}_Z^\top = \boldsymbol{P}_Z$ and $\boldsymbol{P}_Z^2 = \boldsymbol{P}_Z$
  - Thus, $\boldsymbol{P}_Z(\boldsymbol{P}_Z - \boldsymbol{I}_n) = \boldsymbol{0}_{n \times n}$

Two Stage Least Squares: Second Stage

- Regress $\boldsymbol{Y}$ on the first-stage fitted values $\widehat{\boldsymbol{X}}$:

$$\widehat{\boldsymbol{\beta}}_{\text{IV}} = \underset{\boldsymbol{\beta} \in \mathbb{R}^K}{\arg\min} \, (\boldsymbol{Y} - \widehat{\boldsymbol{X}}\boldsymbol{\beta})^\top (\boldsymbol{Y} - \widehat{\boldsymbol{X}}\boldsymbol{\beta})$$

- The first-order condition is

$$-2\widehat{\boldsymbol{X}}^\top \boldsymbol{Y} + 2\widehat{\boldsymbol{X}}^\top \widehat{\boldsymbol{X}} \widehat{\boldsymbol{\beta}}_{\text{IV}} = \boldsymbol{0}_K$$

$$\widehat{\boldsymbol{X}}^\top \widehat{\boldsymbol{X}} \widehat{\boldsymbol{\beta}}_{\text{IV}} = \widehat{\boldsymbol{X}}^\top \boldsymbol{Y}$$

- Therefore, the 2SLS estimator is

$$\widehat{\boldsymbol{\beta}}_{\text{IV}} = (\widehat{\boldsymbol{X}}^\top \widehat{\boldsymbol{X}})^{-1} \widehat{\boldsymbol{X}}^\top \boldsymbol{Y}$$

  which is equivalent to

$$\widehat{\boldsymbol{\beta}}_{\text{IV}} = (\boldsymbol{X}^\top \boldsymbol{P}_Z \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{P}_Z \boldsymbol{Y}$$

Why 2SLS works?

- $\hat{\beta}_{\mathrm{IV}}$ gives β in the linear model. Let's derive why it is the case.
- Substituting $\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{\epsilon}$, we have

$$\begin{aligned}\widehat{\beta}_{\mathrm{IV}} - \beta &= (\boldsymbol{X}^\top \boldsymbol{P}_Z \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{P}_Z \boldsymbol{\epsilon} \\ &= \left(\frac{\boldsymbol{X}^\top \boldsymbol{P}_Z \boldsymbol{X}}{n}\right)^{-1} \frac{\boldsymbol{X}^\top \boldsymbol{Z}}{n} \left(\frac{\boldsymbol{Z}^\top \boldsymbol{Z}}{n}\right)^{-1} \frac{\boldsymbol{Z}^\top \boldsymbol{\epsilon}}{n}.\end{aligned}$$

- By instrument exogeneity and the law of large numbers,

$$\frac{\boldsymbol{Z}^\top \boldsymbol{\epsilon}}{n} = \frac{1}{n} \sum_{i=1}^{n} \boldsymbol{Z}_i \epsilon_i \xrightarrow{p} \mathbb{E}[\boldsymbol{Z}_i \epsilon_i] = \boldsymbol{0}_L.$$

- If $\mathbb{E}[\boldsymbol{Z}_i \boldsymbol{Z}_i^\top]$ is nonsingular and $\operatorname{rank}(\mathbb{E}[\boldsymbol{Z}_i \boldsymbol{X}_i^\top]) = K$, the other factors have finite limits. Therefore,

$$\widehat{\beta}_{\mathrm{IV}} - \beta \xrightarrow{p} \boldsymbol{0}_K \qquad \Longrightarrow \qquad \widehat{\beta}_{\mathrm{IV}} \xrightarrow{p} \beta.$$

Weak Instrument (1)

- Go back to the linear model world:

$$Y_i = \alpha + \beta T_i + \epsilon_i$$
$$T_i = \gamma Z_i + \eta_i$$

  where $\mathbb{E}[\epsilon_i \mid Z_i] = \mathbb{E}[\eta_i \mid Z_i] = 0$

- IV is unstable when instrument weakly affects treatment $\gamma \approx 0$
  - Let's see how bias appears
  - For the sake of simplicity, assume $\bar{Z} = 0$

- Then, Wald estimator is written as

$$\hat{\beta}_{\text{IV}} := \frac{\widehat{\text{Cov}(Y_i, Z_i)}}{\widehat{\text{Cov}(T_i, Z_i)}} = \frac{\frac{1}{n}\sum_{i=1}^{n} Y_i Z_i}{\frac{1}{n}\sum_{i=1}^{n} T_i Z_i}$$

Weak Instrument (2)

$$\hat{\beta}_{\text{IV}} - \beta = \frac{\frac{1}{n}\sum_{i=1}^{n} Y_i Z_i}{\frac{1}{n}\sum_{i=1}^{n} T_i Z_i} - \beta = \frac{\frac{1}{n}\sum_{i=1}^{n}(\alpha + \beta T_i + \epsilon_i) Z_i}{\frac{1}{n}\sum_{i=1}^{n} T_i Z_i} - \beta$$
$$= \beta + \frac{\frac{1}{n}\sum_{i=1}^{n} \epsilon_i Z_i}{\frac{1}{n}\sum_{i=1}^{n} T_i Z_i} - \beta = \frac{\frac{1}{n}\sum_{i=1}^{n} \epsilon_i Z_i}{\frac{1}{n}\sum_{i=1}^{n} T_i Z_i}$$

- If $\gamma = 0$, then $T_i = \gamma Z_i + \eta_i = \eta_i$. So, $\hat{\beta}_{\text{IV}} - \beta = \frac{\frac{1}{n}\sum_{i=1}^{n} \epsilon_i Z_i}{\frac{1}{n}\sum_{i=1}^{n} \eta_i Z_i}$.
- Notice that both numerator and denominator converges to 0

$$\lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \epsilon_i Z_i = \mathbb{E}[\epsilon_i Z_i] = 0, \quad \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} T_i Z_i = \mathbb{E}[\eta_i Z_i] = 0$$

but $0/0$ is undefined, which causes bias

  - By CLT, both numerator and denominator converges to normal distribution. So, the ratio converges to Cauchy distribution, which has no mean and variance.

Anderson–Rubin Test (1)

- Anderson and Rubin (1949): Test $H_0 : \beta = \beta_0$
  - β_0 is the value specified by the null hypothesis

- Subtract $\beta_0 T_i$ from both sides:

$$\begin{aligned}\mathbb{E}[Y_i - \beta_0 T_i \mid Z_i] &= \mathbb{E}[\alpha + (\beta - \beta_0) T_i + \epsilon_i \mid Z_i] \\ &= \mathbb{E}[\alpha + \epsilon_i \mid Z_i] = \alpha \qquad \text{(under } H_0\text{)}\end{aligned}$$

- Regress $Y_i - \beta_0 T_i$ on an intercept and Z_i, and test whether the coefficient on Z_i is zero
  - This argument holds even when $\gamma = 0$

Anderson–Rubin Test (2)

- In matrix notation, first consider a joint null on **all** coefficients, including the intercept:

$$H_0 : \beta = \beta_0 \qquad \Longrightarrow \qquad \boldsymbol{Y} - \boldsymbol{X}\beta_0 = \boldsymbol{\epsilon}$$

- Regress $\boldsymbol{Y} - \boldsymbol{X}\beta_0$ on $\boldsymbol{Z}$

$$\begin{aligned} \text{Fitted values} &= \boldsymbol{P}_Z(\boldsymbol{Y} - \boldsymbol{X}\beta_0) \\ \text{Residuals} &= (\boldsymbol{I}_n - \boldsymbol{P}_Z)(\boldsymbol{Y} - \boldsymbol{X}\beta_0) \end{aligned}$$

- The F-statistic for all L coefficients being zero is

$$F(\beta_0) = \frac{(\boldsymbol{Y} - \boldsymbol{X}\beta_0)^\top \boldsymbol{P}_Z(\boldsymbol{Y} - \boldsymbol{X}\beta_0)/L}{(\boldsymbol{Y} - \boldsymbol{X}\beta_0)^\top (\boldsymbol{I}_n - \boldsymbol{P}_Z)(\boldsymbol{Y} - \boldsymbol{X}\beta_0)/(n-L)}$$

where $\text{rank}(\boldsymbol{Z}) = L < n$

Anderson–Rubin Test (3)

- Under H_0, substitute $\boldsymbol{Y} - \boldsymbol{X}\beta_0 = \epsilon$:

$$F(\beta_0) = \frac{\epsilon^\top \boldsymbol{P}_Z \epsilon / L}{\epsilon^\top (\boldsymbol{I}_n - \boldsymbol{P}_Z)\epsilon / (n - L)}$$

- Suppose $\epsilon \mid \boldsymbol{Z} \sim \mathcal{N}(\boldsymbol{0}_n, \sigma^2 \boldsymbol{I}_n)$. Then, conditional on $\boldsymbol{Z}$,

$$\frac{\epsilon^\top \boldsymbol{P}_Z \epsilon}{\sigma^2} \sim \chi^2_L,$$
$$\frac{\epsilon^\top (\boldsymbol{I}_n - \boldsymbol{P}_Z)\epsilon}{\sigma^2} \sim \chi^2_{n-L}$$

  These are independent because $\boldsymbol{P}_Z(\boldsymbol{I}_n - \boldsymbol{P}_Z) = \boldsymbol{0}$

- Therefore, $F(\beta_0) \sim F_{L,n-L}$
  - Important: The null distribution does not depend on first-stage strength
  - We can invert the test to obtain a confidence interval on $\hat{\beta}$

Review Questions: Setting

- Return to a binary encouragement Z_i, binary treatment T_i, and outcome Y_i.
- Let $T_i(z)$ be treatment received under encouragement z. Write $Y_i(z) := Y_i(z, T_i(z))$ for the outcome under that encouragement, including its effect on treatment.

$$T_i = Z_i T_i(1) + (1 - Z_i) T_i(0), \qquad Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

- As in Module 4, potential-outcome vectors are i.i.d. draws from a super-population, with finite second moments.
- Complete randomization assigns a fixed n_1 units to $Z_i = 1$ and $n_0 = n - n_1$ to $Z_i = 0$, independently of all potential outcomes. Assume $n_1, n_0 \geq 2$. For asymptotics, let $n_1/n \to p \in (0, 1)$.

Review Questions: ITT and CATE

- Define the intention-to-treat effects on outcome and treatment:

$$\mathrm{ITT}_Y = \mathbb{E}[Y_i(1) - Y_i(0)],$$
$$\mathrm{ITT}_T = \mathbb{E}[T_i(1) - T_i(0)].$$

- Randomization identifies both effects as differences in means between encouragement arms. If $\mathrm{ITT}_T \neq 0$,

$$\hat{\beta}_{\mathrm{IV}} = \frac{\widehat{\mathrm{ITT}}_Y}{\widehat{\mathrm{ITT}}_T} \xrightarrow{p} \beta_{\mathrm{IV}} := \frac{\mathrm{ITT}_Y}{\mathrm{ITT}_T}.$$

- With exclusion and monotonicity, this ratio is the **complier average treatment effect (CATE or LATE)**: the treatment effect among units with $T_i(1) = 1, T_i(0) = 0$.

Review Question 1(a)

- The Wald estimator of the complier average treatment effect (CATE, also called LATE) is

$$\hat{\beta}_{\mathrm{IV}} = \frac{\widehat{\mathrm{ITT}}_Y}{\widehat{\mathrm{ITT}}_T}.$$

- Here $\widehat{\mathrm{ITT}}_Y$ and $\widehat{\mathrm{ITT}}_T$ are the coefficients on Z_i from regressing Y_i and T_i, respectively, on an intercept and Z_i.

- Under the IV assumptions and $\mathrm{ITT}_T \neq 0$, this estimator is consistent for $\beta_{\mathrm{IV}} = \mathrm{ITT}_Y / \mathrm{ITT}_T$.

- **Question**: Derive its asymptotic variance using the Delta method.

Review Question 1(a) - Hint: Delta Method

- Suppose $\boldsymbol{X}_n \xrightarrow{p} \boldsymbol{x}$ and

$$\sqrt{n}(\boldsymbol{X}_n - \boldsymbol{x}) \xrightarrow{d} N(\boldsymbol{0}, \Sigma).$$

- For a differentiable function f, let $J(\boldsymbol{x})$ be its Jacobian evaluated at $\boldsymbol{x}$. Then

$$\sqrt{n}\{f(\boldsymbol{X}_n) - f(\boldsymbol{x})\} \xrightarrow{d} N(0, J(\boldsymbol{x})\Sigma J(\boldsymbol{x})^\top).$$

- $J(\boldsymbol{x})\Sigma J(\boldsymbol{x})^\top$ is the variance of the $\sqrt{n}$-scaled limiting distribution. The corresponding first-order sampling variance of $f(\boldsymbol{X}_n)$ is

$$\frac{1}{n} J(\boldsymbol{x})\Sigma J(\boldsymbol{x})^\top.$$

Review Question 1(a) - Hint: An Example

- If $X_1, \ldots, X_n$ are i.i.d. with mean $\mu \neq 0$ and variance σ^2, the LLN and CLT give

$$\bar{X}_n \xrightarrow{p} \mu, \qquad \sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

- To study $\bar{X}_n^2$, take $f(x) = x^2$, so $J(\mu) = f'(\mu) = 2\mu$.

$$\sqrt{n}(\bar{X}_n^2 - \mu^2) \xrightarrow{d} N(0, 4\mu^2\sigma^2).$$

- The limiting variance is $4\mu^2\sigma^2$; the first-order sampling variance is $4\mu^2\sigma^2/n$.

Review Question 1(a) - Answer (1)

- Write the two consistent ITT estimators as a vector:

$$\boldsymbol{X}_n = \begin{bmatrix} \widehat{\mathrm{ITT}}_Y \\ \widehat{\mathrm{ITT}}_T \end{bmatrix}, \qquad \boldsymbol{x} = \begin{bmatrix} \mathrm{ITT}_Y \\ \mathrm{ITT}_T \end{bmatrix}.$$

- With i.i.d. sampled units, finite second moments, and $n_1/n \to p \in (0,1)$, the joint CLT gives

$$\sqrt{n}(\boldsymbol{X}_n - \boldsymbol{x}) \xrightarrow{d} N(\boldsymbol{0}, \Sigma), \qquad \Sigma = \begin{bmatrix} \Sigma_{YY} & \Sigma_{YT} \\ \Sigma_{YT} & \Sigma_{TT} \end{bmatrix}.$$

- The ratio is a differentiable function of this vector:

$$f(a,b) = \frac{a}{b}, \quad f(\boldsymbol{X}_n) = \hat{\beta}_{\mathrm{IV}}, \quad f(\boldsymbol{x}) = \beta_{\mathrm{IV}}.$$

Review Question 1(a) - Answer (2)

- Differentiate $f(a, b) = a/b$ with respect to both arguments:

$$\frac{\partial f}{\partial a} = \frac{1}{b}, \qquad \frac{\partial f}{\partial b} = -\frac{a}{b^2}.$$

- Evaluate at $a = \mathrm{ITT}_Y$, $b = \mathrm{ITT}_T$:

$$J(\boldsymbol{x}) = \begin{bmatrix} \dfrac{1}{\mathrm{ITT}_T} & -\dfrac{\mathrm{ITT}_Y}{\mathrm{ITT}_T^2} \end{bmatrix}.$$

- This linearizes the estimation error:

$$\hat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}} \approx \frac{\widehat{\mathrm{ITT}}_Y - \mathrm{ITT}_Y}{\mathrm{ITT}_T} - \frac{\mathrm{ITT}_Y}{\mathrm{ITT}_T^2}(\widehat{\mathrm{ITT}}_T - \mathrm{ITT}_T).$$

Review Question 1(a) - Answer (3)

- Apply the Delta method, writing $a = \mathrm{ITT}_Y$ and $b = \mathrm{ITT}_T$ to keep the multiplication readable:

$$\begin{aligned} V_\infty &= J(\boldsymbol{x})\Sigma J(\boldsymbol{x})^\top \\ &= \begin{bmatrix} 1/b & -a/b^2 \end{bmatrix} \begin{bmatrix} \Sigma_{YY} & \Sigma_{YT} \\ \Sigma_{YT} & \Sigma_{TT} \end{bmatrix} \begin{bmatrix} 1/b \\ -a/b^2 \end{bmatrix} \\ &= \frac{\Sigma_{YY}}{b^2} + \frac{a^2\Sigma_{TT}}{b^4} - \frac{2a\Sigma_{YT}}{b^3} \\ &= \frac{b^2\Sigma_{YY} + a^2\Sigma_{TT} - 2ab\Sigma_{YT}}{b^4}. \end{aligned}$$

- Therefore,

$$\sqrt{n}(\hat{\beta}_{\mathrm{IV}} - \beta_{\mathrm{IV}}) \xrightarrow{d} N(0, V_\infty).$$

Review Question 1(a) - Answer (4)

- Divide the limiting variance by n to obtain the usual first-order sampling variance formula:

$$V_{\text{Wald},n} \approx \frac{1}{\text{ITT}_T^4}\Big\{\text{ITT}_T^2\mathbb{V}(\widehat{\text{ITT}}_Y) + \text{ITT}_Y^2\mathbb{V}(\widehat{\text{ITT}}_T) - 2\text{ITT}_Y\text{ITT}_T\,\text{Cov}(\widehat{\text{ITT}}_Y, \widehat{\text{ITT}}_T)\Big\}.$$

- This uses $\Sigma_{YY}/n \approx \mathbb{V}(\widehat{\text{ITT}}_Y)$, $\Sigma_{TT}/n \approx \mathbb{V}(\widehat{\text{ITT}}_T)$, and $\Sigma_{YT}/n \approx \text{Cov}(\widehat{\text{ITT}}_Y, \widehat{\text{ITT}}_T)$.

- Estimate this expression by plugging in estimated ITTs, variances, and covariance. Its square root is the Wald standard error.

Review Question 1(b) (Optional)

- Under complete randomization of Z_i and simple random sampling of units, show that

$$\mathrm{Cov}(\widehat{\mathrm{ITT}}_Y, \widehat{\mathrm{ITT}}_T) = \frac{\mathrm{Cov}(Y_i(1), T_i(1))}{n_1} + \frac{\mathrm{Cov}(Y_i(0), T_i(0))}{n_0}.$$

- Here $Y_i(z) = Y_i(Z_i = z, T_i(z))$ is the outcome under encouragement z, and $T_i(z)$ is treatment received under that encouragement. The sampled potential-outcome vectors are i.i.d.

- **Hint**: Binary-regressor OLS coefficients are differences in means. Condition on the assignment vector $\boldsymbol{Z}$.

Review Question 1(b) - Answer (1)

- For $z \in \{0, 1\}$, define the observed group means

$$\bar{Y}_z = \frac{1}{n_z} \sum_{i:Z_i=z} Y_i(z), \qquad \bar{T}_z = \frac{1}{n_z} \sum_{i:Z_i=z} T_i(z).$$

  Then $\widehat{\mathrm{ITT}}_Y = \bar{Y}_1 - \bar{Y}_0$ and $\widehat{\mathrm{ITT}}_T = \bar{T}_1 - \bar{T}_0$.

- Expand using bilinearity of covariance:

$$\begin{aligned} &\mathrm{Cov}(\widehat{\mathrm{ITT}}_Y, \widehat{\mathrm{ITT}}_T) \\ &= \mathrm{Cov}(\bar{Y}_1, \bar{T}_1) - \mathrm{Cov}(\bar{Y}_1, \bar{T}_0) \\ &\quad - \mathrm{Cov}(\bar{Y}_0, \bar{T}_1) + \mathrm{Cov}(\bar{Y}_0, \bar{T}_0). \end{aligned}$$

- We will find the same-group terms and show that the two cross-group terms vanish.

Review Question 1(b) - Answer (2)

- Conditional on $\boldsymbol{Z}$, the membership of each group is fixed. Randomization leaves the distribution of potential outcomes unchanged:

$$\begin{aligned}\mathrm{Cov}(\bar{Y}_z, \bar{T}_z \mid \boldsymbol{Z}) &= \frac{1}{n_z^2} \sum_{i:Z_i=z} \sum_{j:Z_j=z} \mathrm{Cov}(Y_i(z), T_j(z) \mid \boldsymbol{Z}) \\ &= \frac{1}{n_z^2} \sum_{i:Z_i=z} \mathrm{Cov}(Y_i(z), T_i(z)) \\ &= \frac{\mathrm{Cov}(Y_i(z), T_i(z))}{n_z}.\end{aligned}$$

- I.i.d. sampling makes all $i \neq j$ terms zero. There are n_z identical $i = j$ terms, each allowing dependence between a unit's outcome and treatment.

Review Question 1(b) - Answer (3)

- The two encouragement groups contain different units. Thus,

$$\begin{aligned}\mathrm{Cov}(\bar{Y}_1, \bar{T}_0 \mid \boldsymbol{Z}) &= \frac{1}{n_1 n_0} \sum_{i:Z_i=1} \sum_{j:Z_j=0} \mathrm{Cov}(Y_i(1), T_j(0) \mid \boldsymbol{Z}) \\ &= 0,\end{aligned}$$

  because $i \neq j$ in every term. Similarly, $\mathrm{Cov}(\bar{Y}_0, \bar{T}_1 \mid \boldsymbol{Z}) = 0$.

- To move from conditional to unconditional covariance, use

$$\begin{aligned}\mathrm{Cov}(A, B) = \mathbb{E}[\mathrm{Cov}(A, B \mid \boldsymbol{Z})] \\ + \mathrm{Cov}(\mathbb{E}[A \mid \boldsymbol{Z}], \mathbb{E}[B \mid \boldsymbol{Z}]).\end{aligned}$$

- Here $\mathbb{E}[\bar{Y}_z \mid \boldsymbol{Z}] = \mathbb{E}[Y_i(z)]$ and $\mathbb{E}[\bar{T}_z \mid \boldsymbol{Z}] = \mathbb{E}[T_i(z)]$ are constants, so the second term is zero for every pair of group means.

Review Question 1(b) - Answer (4)

- Substitute the two same-group terms and the zero cross-group terms:

$$\begin{aligned}&\mathrm{Cov}(\widehat{\mathrm{ITT}}_Y, \widehat{\mathrm{ITT}}_T)\\ &= \frac{\mathrm{Cov}(Y_i(1), T_i(1))}{n_1} + \frac{\mathrm{Cov}(Y_i(0), T_i(0))}{n_0}.\end{aligned}$$

- Estimate the covariance in each group by its sample covariance:

$$\widehat{C} = \frac{s_{YT,1}}{n_1} + \frac{s_{YT,0}}{n_0}, \qquad s_{YT,z} = \frac{\sum_{i:Z_i=z}(Y_i - \bar{Y}_z)(T_i - \bar{T}_z)}{n_z - 1}.$$

- This supplies the covariance term needed for the Wald standard error in part (a). The result averages over both random sampling and assignment.

Review Question 2

- Replicate the lecture's GOTV application, assuming monotonicity and the other IV assumptions.

- Estimate the proportions of compliers, always-takers, and never-takers, together with their standard errors.

- Estimate the complier average treatment effect (CATE) on turnout in the subsequent election in two ways:
  1. The Wald estimator and its asymptotic variance.
  2. Two-stage least squares and its heteroskedasticity-robust variance. You may use `AER::ivreg()`.

Review Question 2: Data

- Use `GOTVdata.csv` from the lecture's data directory:
  - Z_i: `treat3`, randomized attempted GOTV contact during the August 2006 primary election.
  - T_i: `AUG2006`, turnout in that election.
  - Y_i: `AUG2008`, turnout in the August 2008 primary.

```
# Run from the directory containing Module5.tex
d <- read.csv(paste0("../../../../lectures/",
  "instrumental_variables/data/GOTVdata.csv"))
Z <- d$treat3
T <- d$AUG2006
Y <- d$AUG2008
n1 <- sum(Z == 1)
n0 <- sum(Z == 0)
```

Review Question 2 - Answer (1)

- Let p_C, p_A, p_N denote the proportions of compliers, always-takers, and never-takers.
- Randomization, consistency, and monotonicity imply

$$\mathbb{E}[T_i \mid Z_i = 0] = \mathbb{P}(T_i(0) = 1) = p_A,$$
$$\mathbb{E}[T_i \mid Z_i = 1] = \mathbb{P}(T_i(1) = 1) = p_A + p_C.$$

- Thus, writing $\bar{T}_z$ for the mean in encouragement arm z,

$$\hat{p}_A = \bar{T}_0, \qquad \hat{p}_C = \bar{T}_1 - \bar{T}_0, \qquad \hat{p}_N = 1 - \bar{T}_1.$$

- These three estimates sum to one. Only the complier proportion is a difference in means.

Review Question 2 - Answer (2)

- Let s^2_{Tz} be the sample variance of T_i among units with $Z_i = z$, using denominator $n_z - 1$.
- Under random sampling and complete randomization, the estimated variances are

$$\widehat{\mathbb{V}}(\hat{p}_A) = \frac{s^2_{T0}}{n_0},$$

$$\widehat{\mathbb{V}}(\hat{p}_N) = \frac{s^2_{T1}}{n_1},$$

$$\widehat{\mathbb{V}}(\hat{p}_C) = \frac{s^2_{T1}}{n_1} + \frac{s^2_{T0}}{n_0}.$$

- Take square roots to obtain standard errors. The last expression is the usual Neyman variance estimator for the first stage.

Review Question 2 - Answer (3)

- Estimate each stratum proportion and its standard error in R:

```
pA <- mean(T[Z == 0])
pN <- 1 - mean(T[Z == 1])
pC <- mean(T[Z == 1]) - mean(T[Z == 0])

vT0 <- var(T[Z == 0]) / n0
vT1 <- var(T[Z == 1]) / n1
seA <- sqrt(vT0)
seN <- sqrt(vT1)
seC <- sqrt(vT1 + vT0)

c(complier = pC, always = pA, never = pN)
c(complier = seC, always = seA, never = seN)
```

Review Question 2 - Answer (4)

- There are $n_1 = 36{,}893$ encouraged units and $n_0 = 184{,}749$ control units.

Principal stratum	Estimate (%)	SE (percentage points)
Compliers	8.30	0.28
Always-takers	31.13	0.11
Never-takers	60.58	0.25

- Most units would not vote in 2006 under either encouragement assignment; about 31% would vote under either assignment.

- About 8.3% would vote in 2006 if encouraged, but would not vote without encouragement. The IV effect applies to this group.

Review Question 2 - Answer (5)

- Compute the two ITT estimates and their estimated variances:

```
ITT_Y <- mean(Y[Z == 1]) - mean(Y[Z == 0])
ITT_T <- mean(T[Z == 1]) - mean(T[Z == 0])

VarITT_Y <- var(Y[Z == 1]) / n1 +
  var(Y[Z == 0]) / n0
VarITT_T <- vT1 + vT0

CovTerm <- cov(Y[Z == 1], T[Z == 1]) / n1 +
  cov(Y[Z == 0], T[Z == 0]) / n0
```

- The covariance term follows from Question 1(b). The estimates are $\widehat{\mathrm{ITT}}_Y = 0.01037$ and $\widehat{\mathrm{ITT}}_T = 0.08296$.

Review Question 2 - Answer (6)

- The Wald estimator divides the reduced form by the first stage:

$$\hat{\beta}_{\mathrm{IV}} = \frac{\widehat{\mathrm{ITT}}_Y}{\widehat{\mathrm{ITT}}_T}.$$

- Plug the estimated ITTs, variances, and covariance into the Delta-method expression from Question 1(a):

$$\begin{aligned}\widehat{V}_{\mathrm{Wald}} = \frac{1}{\widehat{\mathrm{ITT}}_T^4}\Big\{&\widehat{\mathrm{ITT}}_T^2\,\widehat{\mathbb{V}}(\widehat{\mathrm{ITT}}_Y) \\ &+ \widehat{\mathrm{ITT}}_Y^2\,\widehat{\mathbb{V}}(\widehat{\mathrm{ITT}}_T) \\ &- 2\widehat{\mathrm{ITT}}_Y\widehat{\mathrm{ITT}}_T\,\widehat{\mathrm{Cov}}(\widehat{\mathrm{ITT}}_Y, \widehat{\mathrm{ITT}}_T)\Big\}.\end{aligned}$$

- The large-sample 95% confidence interval is $\hat{\beta}_{\mathrm{IV}} \pm 1.96\sqrt{\widehat{V}_{\mathrm{Wald}}}$.

Review Question 2 - Answer (7)

- Compute the Wald estimate, standard error, and interval:

```
estCATE_wald <- ITT_Y / ITT_T

waldVariance <- (ITT_T^2 * VarITT_Y +
  ITT_Y^2 * VarITT_T -
  2 * ITT_Y * ITT_T * CovTerm) / ITT_T^4

se_wald <- sqrt(waldVariance)
CI_wald <- estCATE_wald + c(-1, 1) *
  1.96 * se_wald

c(estimate = estCATE_wald, SE = se_wald)
CI_wald
```

Review Question 2 - Answer (8)

- With an intercept, one binary instrument, and no covariates, 2SLS gives the same point estimate as the Wald ratio:

```
ivmodel <- AER::ivreg(
  AUG2008 ~ AUG2006 | treat3, data = d)

ivtable <- coef(summary(ivmodel,
  vcov = sandwich::sandwich, df = Inf))
estCATE_iv <- ivtable["AUG2006", "Estimate"]
se_iv <- ivtable["AUG2006", "Std. Error"]

all.equal(unname(estCATE_iv), estCATE_wald)
```

- Use the IV sandwich variance, which is heteroskedasticity-robust. The usual second-stage OLS standard error is not valid for 2SLS.

Review Question 2 - Answer (9)

Method	CATE estimate	Standard error
Wald (Delta method)	0.1250	0.03155
2SLS (robust)	0.1250	0.03155

- The Wald 95% confidence interval is [0.0632, 0.1868]. The two standard errors agree to the reported precision; finite-sample variance conventions differ slightly.

- Under the IV assumptions, voting in 2006 increases the probability of voting in 2008 by about **12.5 percentage points for compliers**.

- The interval excludes zero. This is a local effect for voters whose 2006 turnout was changed by encouragement.

Review Question 3

- Derive the asymptotic bias of the Wald estimator when:
  (a) The **exclusion restriction fails**, while randomization and monotonicity hold.

  (b) **Monotonicity fails**, while randomization and the exclusion restriction hold.

- **Hint**: revisit the principal strata. What changes in the numerator and denominator of

$$\widehat{\beta}_{\text{Wald}} = \frac{\widehat{\text{ITT}}_Y}{\widehat{\text{ITT}}_T} \xrightarrow{p} \beta_{\text{IV}} := \frac{\text{ITT}_Y}{\text{ITT}_T}?$$

- Assume a nonzero first stage, $\text{ITT}_T \neq 0$.

Review Question 3 - Answer (a1)

- Without exclusion, retain both encouragement and treatment in the potential outcome. Define the encouragement effect

$$\Delta_i := Y_i(1, T_i(1)) - Y_i(0, T_i(0)).$$

- Monotonicity leaves compliers C and non-compliers NC (always-takers and never-takers). Write

$$p_C := \mathbb{P}(C), \quad p_{NC} := 1 - p_C, \qquad \delta_s := \mathbb{E}[\Delta_i \mid s], \quad s \in \{C, NC\}.$$

- By the law of iterated expectation and randomization,

$$\mathrm{ITT}_Y = \mathbb{E}[\Delta_i] = p_C \delta_C + p_{NC} \delta_{NC}.$$

- Exclusion would imply $\delta_{NC} = 0$; it need not be zero here.

Review Question 3 - Answer (a2)

- Monotonicity still identifies the complier proportion:

$$\mathrm{ITT}_T = \mathbb{E}[T_i(1) - T_i(0)] = 1 \times p_C + 0 \times p_{NC} = p_C > 0.$$

- Substitute the numerator and denominator:

$$\begin{aligned}\beta_{\mathrm{IV}} &= \frac{p_C \delta_C + p_{NC} \delta_{NC}}{p_C} \\ &= \delta_C + \delta_{NC} \frac{p_{NC}}{p_C}.\end{aligned}$$

- Thus, the asymptotic discrepancy from the complier ITT is

$$\beta_{\mathrm{IV}} - \delta_C = \delta_{NC} \frac{p_{NC}}{p_C}.$$

The comparison target δ_C is the **complier ITT effect**.

Review Question 3 - Answer (a3)

- The discrepancy grows in magnitude with
  - the encouragement effect on non-compliers, $|\delta_{NC}|$;
  - the ratio of non-compliers to compliers, p_{NC}/p_C.
- Without exclusion, even the complier ITT effect changes both encouragement and treatment:

$$\delta_C = \mathbb{E}[Y_i(1,1) - Y_i(0,0) \mid C].$$

- It need not equal the complier treatment effect at a fixed encouragement level, such as

$$\mathbb{E}[Y_i(0,1) - Y_i(0,0) \mid C].$$

- Consequently, the displayed bias formula alone does not isolate bias relative to a treatment-only CATE when exclusion fails.

Review Question 3 - Answer (b1)

- Now maintain exclusion and allow defiers D:

$$T_i(1) = 0, \qquad T_i(0) = 1.$$

Let $p_s := \mathbb{P}(s)$ for strata $s \in \{C, A, N, D\}$, where A and N denote always-takers and never-takers.

- The first stage becomes

$$\begin{aligned} \mathrm{ITT}_T &= \mathbb{P}(T_i(1) = 1) - \mathbb{P}(T_i(0) = 1) \\ &= (p_C + p_A) - (p_A + p_D) \\ &= p_C - p_D. \end{aligned}$$

- Compliers contribute $+1$, defiers contribute -1, and the other strata contribute zero. Assume $p_C - p_D \neq 0$.

Review Question 3 - Answer (b2)

- Under exclusion, write $Y_i^T(t)$ for the outcome under treatment t. Define treatment effects within each affected stratum:

$$\tau_s := \mathbb{E}[Y_i^T(1) - Y_i^T(0) \mid s], \qquad s \in \{C, D\}.$$

- Encouragement moves compliers into treatment and defiers out of it:

$$\delta_C = \mathbb{E}[\Delta_i \mid C] = \tau_C = \mathrm{CATE},$$
$$\delta_D = \mathbb{E}[\Delta_i \mid D] = -\tau_D.$$

- Always-takers and never-takers have zero encouragement effects, so

$$\mathrm{ITT}_Y = p_C\delta_C + p_D\delta_D = p_C\tau_C - p_D\tau_D,$$
$$\beta_{\mathrm{IV}} = \frac{p_C\delta_C + p_D\delta_D}{p_C - p_D}.$$

Review Question 3 - Answer (b3)

- Subtract the complier treatment effect, $\delta_C = \tau_C$:

$$\begin{aligned}
\beta_{\text{IV}} - \tau_C &= \frac{p_C \delta_C + p_D \delta_D}{p_C - p_D} - \delta_C \\
&= \frac{p_C \delta_C + p_D \delta_D - \delta_C (p_C - p_D)}{p_C - p_D} \\
&= (\delta_C + \delta_D) \frac{p_D}{p_C - p_D} \\
&= (\tau_C - \tau_D) \frac{p_D}{p_C - p_D}.
\end{aligned}$$

- Equivalently, using the defier encouragement effect,

$$\beta_{\text{IV}} - \text{CATE} = (\text{CATE} + \text{ITT}_D) \frac{p_D}{p_C - p_D},$$

where $\text{ITT}_D = \delta_D = -\tau_D$.

Review Question 3 - Answer (b4)

- The asymptotic bias is

$$\beta_{\mathrm{IV}} - \mathrm{CATE} = (\tau_C - \tau_D)\frac{p_D}{p_C - p_D}.$$

- It is zero if there are no defiers, or if compliers and defiers have the same average treatment effect.

- It is small when treatment effects are similar and/or the defier proportion is small relative to the net first stage, $|p_C - p_D|$.

- When p_C and p_D nearly cancel, the denominator is small and even a small defier proportion can produce substantial bias.

Extra: Many-Instrument Bias (1)

- More instruments allow you to test assumptions (overidentification), but they can also introduce bias in 2SLS (many-instrument problem)
- For simplicity, consider one endogenous regressor:

$$\boldsymbol{Y} = \beta \boldsymbol{T} + \boldsymbol{\epsilon}$$
$$\boldsymbol{T} = \boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\eta}$$

  where $\mathbb{E}[\boldsymbol{\epsilon} \mid \boldsymbol{Z}] = \mathbb{E}[\boldsymbol{\eta} \mid \boldsymbol{Z}] = 0$, but $\mathrm{Cov}(\eta_i, \epsilon_i) \neq 0$.

- Recall that the first-stage fitted values are

$$\widehat{\boldsymbol{T}} = \boldsymbol{P}_Z \boldsymbol{T} = \boldsymbol{P}_Z(\boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{\eta}) = \boldsymbol{Z}\boldsymbol{\gamma} + \underbrace{\boldsymbol{P}_Z \boldsymbol{\eta}}_{\text{Estimation error}}$$

- For each observation i, we have

$$\widehat{T}_i = \boldsymbol{Z}_i^\top \boldsymbol{\gamma} + \sum_{j=1}^{n} (\boldsymbol{P}_Z)_{ij} \eta_j = \boldsymbol{Z}_i^\top \boldsymbol{\gamma} + \underbrace{(\boldsymbol{P}_Z)_{ii} \eta_i}_{\text{Own observation}} + \sum_{j \neq i} (\boldsymbol{P}_Z)_{ij} \eta_j$$

Extra: Many-Instrument Bias (2)

- Recall the 2SLS estimator:

$$\widehat{\beta}_{\text{2SLS}} = \frac{\widehat{\boldsymbol{T}}^\top \boldsymbol{Y}}{\widehat{\boldsymbol{T}}^\top \boldsymbol{T}} = \beta + \frac{\widehat{\boldsymbol{T}}^\top \boldsymbol{\epsilon}}{\widehat{\boldsymbol{T}}^\top \boldsymbol{T}}$$

- Substituting the first-stage fitted values:

$$\widehat{\beta}_{\text{2SLS}} - \beta = \frac{(\boldsymbol{Z}\boldsymbol{\gamma} + \boldsymbol{P}_Z\boldsymbol{\eta})^\top \boldsymbol{\epsilon}}{\boldsymbol{T}^\top \boldsymbol{P}_Z \boldsymbol{T}} = \frac{\boldsymbol{\gamma}^\top \boldsymbol{Z}^\top \boldsymbol{\epsilon} + \boldsymbol{\eta}^\top \boldsymbol{P}_Z \boldsymbol{\epsilon}}{\boldsymbol{T}^\top \boldsymbol{P}_Z \boldsymbol{T}}$$

- Notice that the additional term is

$$\boldsymbol{\eta}^\top \boldsymbol{P}_Z \boldsymbol{\epsilon} = \underbrace{\sum_{i=1}^{n} (\boldsymbol{P}_Z)_{ii} \eta_i \epsilon_i}_{\text{Own-observation term}} + \sum_{i \neq j} (\boldsymbol{P}_Z)_{ij} \eta_j \epsilon_i$$

- Since $\text{Cov}(\eta_i, \epsilon_i) \neq 0$, the own-observation term has nonzero expectation!

Extra: Many-Instrument Bias (3)

- Assume independent observations and $\mathbb{E}[\eta_i \epsilon_i \mid \boldsymbol{Z}] = \sigma_{\eta\epsilon} \neq 0$.
- Taking the conditional expectation:

$$\mathbb{E}[\widehat{\boldsymbol{T}}^\top \boldsymbol{\epsilon} \mid \boldsymbol{Z}] = \mathbb{E}[\boldsymbol{\eta}^\top \boldsymbol{P}_Z \boldsymbol{\epsilon} \mid \boldsymbol{Z}] = \sum_{i=1}^{n} (\boldsymbol{P}_Z)_{ii} \sigma_{\eta\epsilon} = \underbrace{\operatorname{tr}(\boldsymbol{P}_Z)}_{=L} \sigma_{\eta\epsilon} = L\sigma_{\eta\epsilon}$$

  where L is the number of instruments (columns of $\boldsymbol{Z}$).

- Dividing by the sample size gives $\mathbb{E}\left[\frac{1}{n}\widehat{\boldsymbol{T}}^\top \boldsymbol{\epsilon} \mid \boldsymbol{Z}\right] = \frac{L}{n}\sigma_{\eta\epsilon}$
- If L is fixed, $L/n \to 0$. But if $L/n \to \alpha > 0$, this term does not vanish.
- This is the source of the many-instrument bias of 2SLS.
- Solution: Split-Sample IV / Jackknife IV (JIVE)

Extra: Split-Sample IV

- Angrist and Krueger (1995): Estimate the first stage using different observations from the second stage
- Randomly split the observations into two halves, denoted by (1) and (2). Use the first half to predict $\boldsymbol{X}$ in the second:

$$\widehat{\boldsymbol{X}}_{(2)} = \boldsymbol{Z}_{(2)}(\boldsymbol{Z}_{(1)}^\top \boldsymbol{Z}_{(1)})^{-1}\boldsymbol{Z}_{(1)}^\top \boldsymbol{X}_{(1)}$$

  Assume the displayed inverses exist

- The original split-sample IV (SSIV) regresses $\boldsymbol{Y}_{(2)}$ on these predictions:

$$\widehat{\boldsymbol{\beta}}_{\text{SSIV}} = (\widehat{\boldsymbol{X}}_{(2)}^\top \widehat{\boldsymbol{X}}_{(2)})^{-1}\widehat{\boldsymbol{X}}_{(2)}^\top \boldsymbol{Y}_{(2)}$$

- Limitation: it does not work well in small samples

Extra: Jacknife-IV (JIVE)

- Angrist, Imbens, and Krueger (1999): Leave observation i out of the first stage when predicting its $\boldsymbol{X}_i$
- Let $\boldsymbol{X}_{(-i)}$ and $\boldsymbol{Z}_{(-i)}$ remove row i. The prediction for that observation is

$$\widehat{\boldsymbol{X}}_{i,(-i)}^{\top} = \boldsymbol{Z}_i^{\top}(\boldsymbol{Z}_{(-i)}^{\top}\boldsymbol{Z}_{(-i)})^{-1}\boldsymbol{Z}_{(-i)}^{\top}\boldsymbol{X}_{(-i)}$$

- With i.i.d. observations and $\mathbb{E}[\epsilon_i \mid \boldsymbol{Z}_i] = 0$,

$$\mathbb{E}[\widehat{\boldsymbol{X}}_{i,(-i)}\epsilon_i \mid \boldsymbol{Z}] = \boldsymbol{0}_K$$

  Observation i's own $\boldsymbol{X}_i$ is not used to fit its prediction

- Stack these predictions in $\widehat{\boldsymbol{X}}_{(-)}$: row i is $\widehat{\boldsymbol{X}}_{i,(-i)}^{\top}$
- Use the leave-one-out predictions as instruments:

$$\widehat{\boldsymbol{X}}_{(-)}^{\top}(\boldsymbol{Y} - \boldsymbol{X}\widehat{\beta}_{\text{JIVE}}) = \boldsymbol{0}_K$$

$$\widehat{\beta}_{\text{JIVE}} = (\widehat{\boldsymbol{X}}_{(-)}^{\top}\boldsymbol{X})^{-1}\widehat{\boldsymbol{X}}_{(-)}^{\top}\boldsymbol{Y}$$